QUIETKEY

Revised Prototype Hardware Blueprint

QK2-HW-P0.1 | BLOCKER-RESOLUTION ISSUE

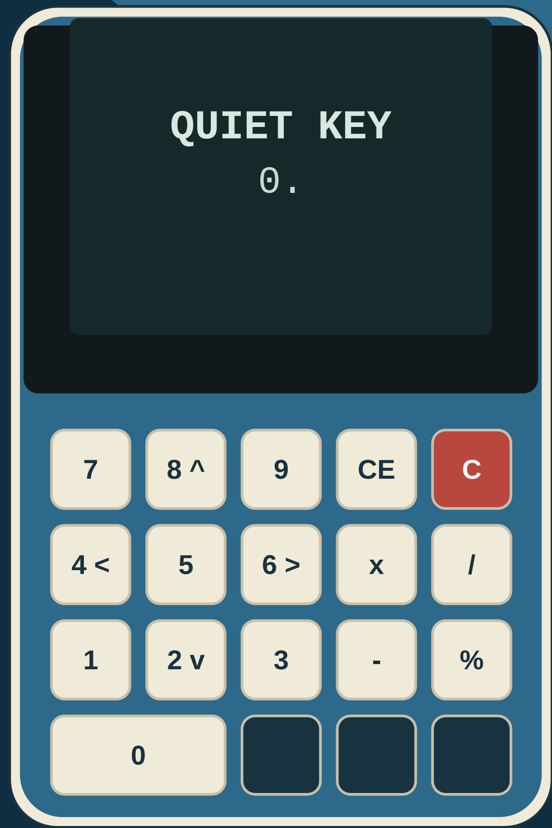
SELECTED PROTOTYPE ENVELOPE

112 x 74 x 20 mm

H x W x maximum D | target mass 165-195 g

- No radio silicon; no external data connector; removable 4 x AAA L92 supply.
- Top-edge ISO 7816 keycard slot; transaction microSD concealed behind battery door.
- No wallet secret at rest; native P2WSH 2-of-2 system basis remains binding.
- Footer-token OCR optics; sealed camera window; card-present-only wallet lighting.

EXPERIMENTAL - PCB AND ENCLOSURE RELEASE BLOCKED



ISSUED 16 AUG 2026

1 BUILD INTENT

This document freezes a prototype architecture that a third-party shop can quote, CAD, fabricate and assemble. It is not a security certification, production release, or substitute for QK2 Gates A-C.

2 BINDING SYSTEM BASIS

QK2-04 remains authoritative: no terminal wallet secret at rest; air-gapped calculator decoy; native P2WSH 2-of-2; physical confirmation; signed microSD update; no radio silicon or external data port.

3 REVIEW RECONCILIATION

RV1 is adopted where technically correct. Its 70 x 110 x 16 QK2-04 baseline and attributed no-card lighting clauses are not present in the supplied QK2-04; the lighting rules are adopted here as new P0.1 decisions, not as quotations from QK2-04.

DECISION REGISTER

Requirement / issue	QK2-04 basis	P0.1 decision	Disposition
Overall envelope	80 x 160 x 18 mm target	112 x 74 x 20 mm	Controlled mechanical deviation; approve in QK2-05 before design freeze.
Display	480 x 320 IPS; active approx. 73 x 49	2.8 in 320 x 240 IPS module; 70.2 x 50.2 board	Controlled deviation. Full addresses shown over two lines with no truncation.
Compute	Linux CM-class SoM or equivalent	CM0, no-wireless, 8 GB eMMC - P0 provisional	Release blocker: procure CM0000008, measure memory headroom, and issue successor note.
Keycard	Side ISO 7816 contact slot	Centered top-edge push-pull slot	Controlled deviation retained; 500-cycle shutter test required.
microSD	Behind battery door	Top of bay, downward-facing push-push socket	Conforms in location; accessible without removing cells.
Camera / reader	IMX708; 142-char footer-token capture	Camera 3, sealed AR window, aperture inserts	QR specification withdrawn. Gate A freezes confidence thresholds and optics.
Wallet legends	No explicit illumination rule in supplied QK2-04	Invisible unlit; LEDs inhibited without card	Stealth improvement; numeric calculator legends remain visible.

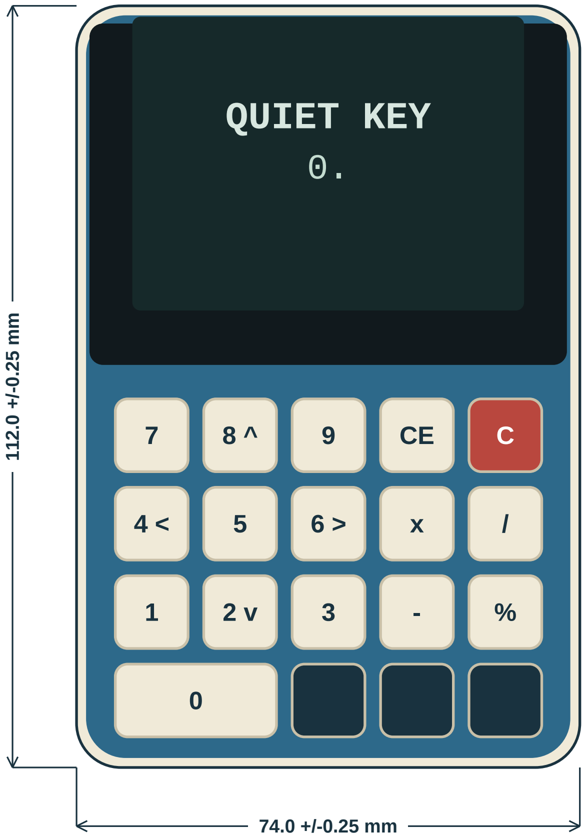
FREEZE RULE **Do not release PCB or enclosure files until owner decisions, optics, CM0 memory, CSI routing and camera-window tests close.**

External envelope and interface placement

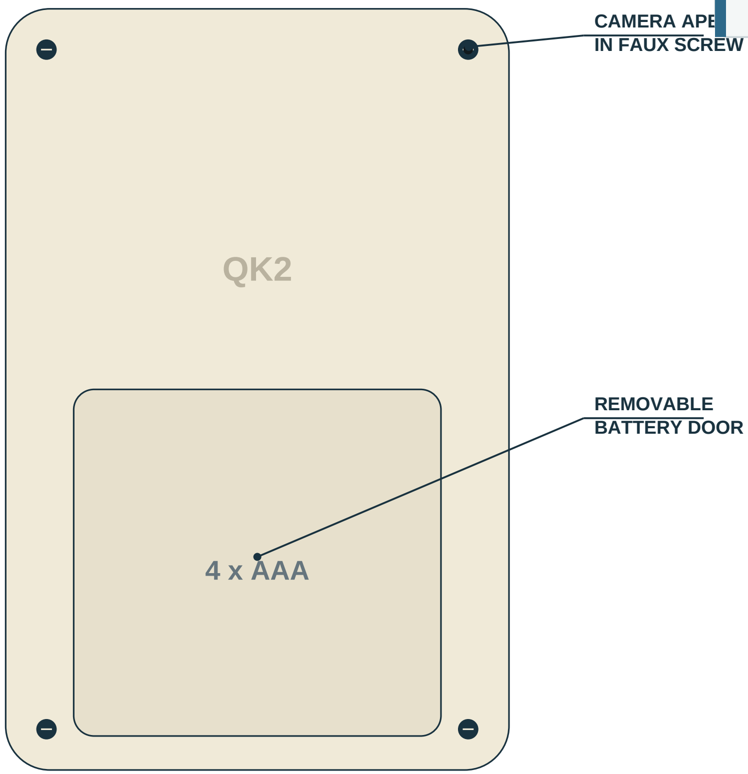
INTERFACE CHOICE

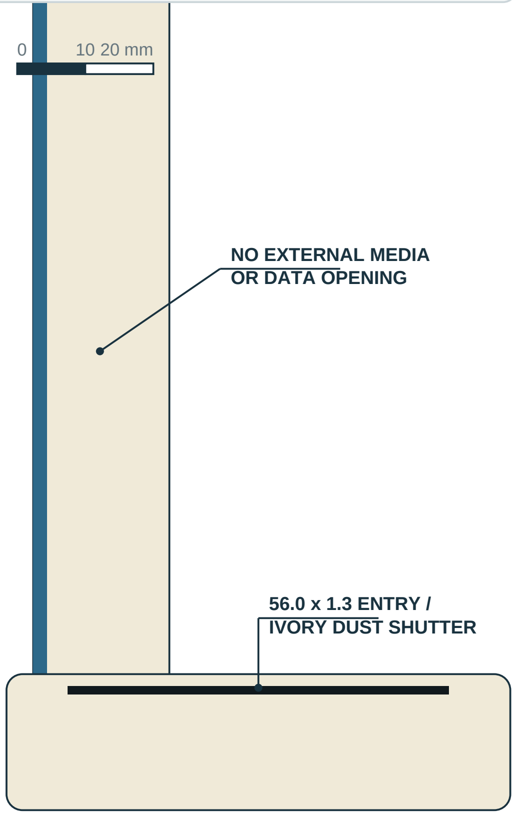
Top-edge card entry keeps the blue face intact and aligns the 55.5 mm connector across the case. Its ivory shutter presents as a seam when empty. microSD is returned behind the battery door: this removes a modern-looking side slot while a downward-facing socket at the top of the bay remains accessible without removing cells.

FRONT



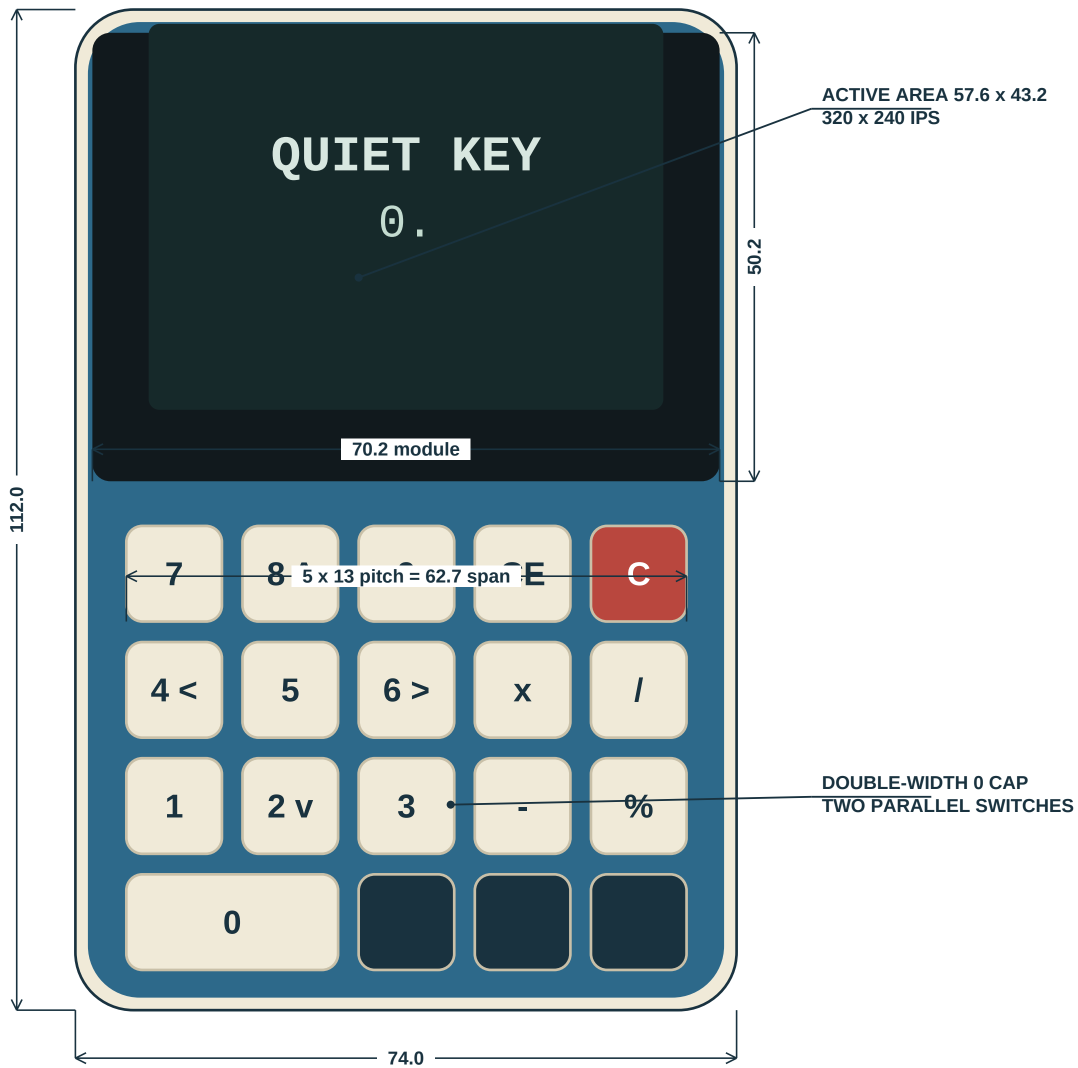
REAR





TOP EDGE - centered card entrance

Front panel and keypad datum plan



1 KEY GEOMETRY

19 visible keycaps over 20 switches. Standard caps: 10.7 x 10.7 mm; pitch 13.0 mm. The 0 cap is 23.7 x 10.7 mm and bridges two mechanically independent, electrically paralleled switches. Minimum cap travel 0.20 mm; target actuation force 1.6-2.2 N at cap center.

2 NAVIGATION MAP

8 = up; 2 = down; 4 = left; 6 = right. Red C = cancel/back and power-on request. CE = delete. Equals = confirm/enter. Secondary wallet legends use translucent ablated cores and are invisible at arm's length when unlit; calculator digits/operators remain visibly printed.

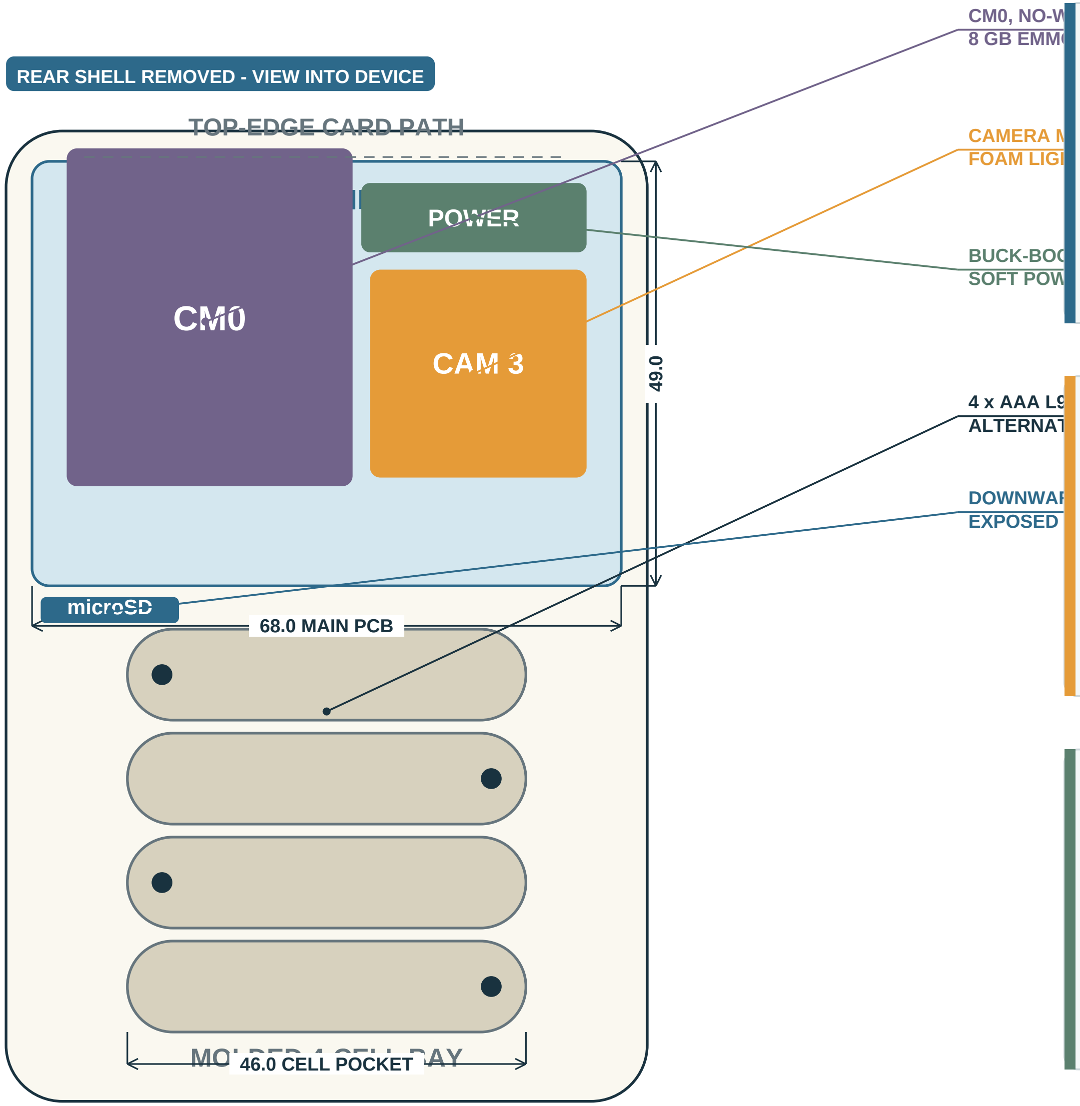
3 FACE CONSTRUCTION

0.8 mm blue anodized 5052-H32 aluminum faceplate bonded only to the removable front carrier - not to the shell. 0.5 mm smoke acrylic display lens. Key wells use a black 1.0 mm printed light fence. No printed Bitcoin or wallet marking on external surfaces.

4 BACKLIGHT TARGET

Neutral white 4000-5000 K; 5-15 cd/m2 at 30% PWM. Hardware inhibit keeps LEDs dark whenever card-detect is inactive. Illumination is wallet-mode only; verify colour coherence beside green and amber display themes.

Open rear: component placement



1 UPPER STACK - 59.5 TO 109 MM DATUM

Main carrier PCB occupies 68 x 49 mm. The 55.5 x 40 x 5.8 mm smartcard connector mounts on its front face, directly behind the display module, with the card entrance at the top. CM0 mounts on the rear face. Camera 3 sits in a 27 x 26 pocket at upper-right and connects by a short 15-pin FPC.

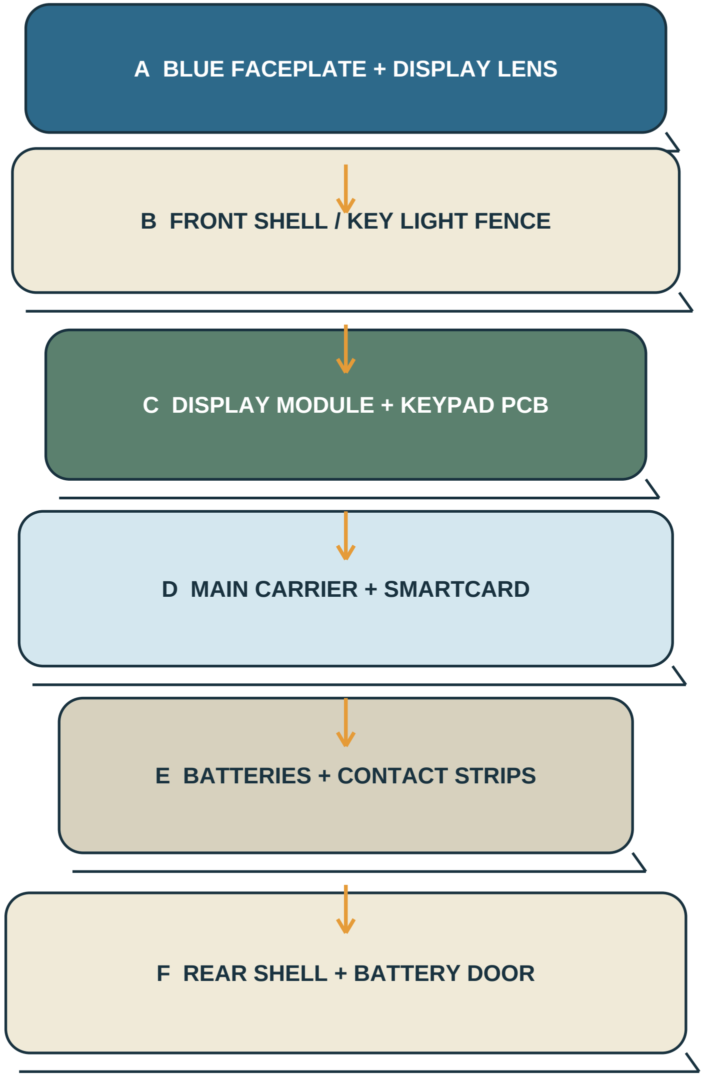
2 LOWER STACK - 5 TO 59 MM DATUM

The rear shell forms four horizontal AAA channels. Cells sit behind the keypad PCB. The microSD mouth is at the top-left of the door opening and accepts the card upward into a connector on the carrier's lower edge; cells remain installed. The 54 x 51 mm door captures cells and provides 0.5-0.8 mm compression pads.

3 CLEARANCE RULES

Keep a 1.0 mm no-component perimeter around both PCBs; 0.5 mm minimum board-to-wall assembly clearance; 1.0 mm between battery metal and PCB solder joints; 0.4 mm FPC bend clearance; 0.8 mm compressible foam around the camera lens; no adhesive over serviceable connectors.

Exploded assembly

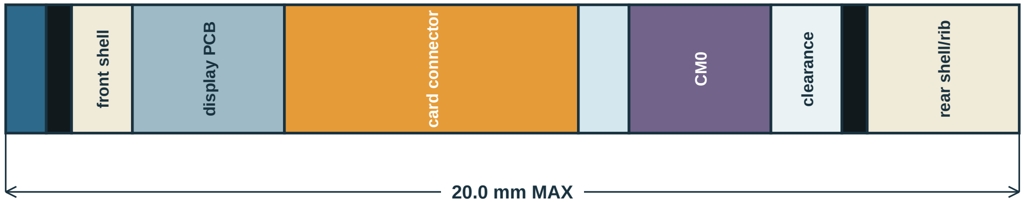


- **A** 0.8 mm Al faceplate; 0.5 mm smoke lens; 19 SLA caps.
- **B** MJF PA12 front shell; molded/printed 1.0 mm light wells.
- **C** 70.2 x 50.2 display board; 68 x 54 keypad PCB.
- **D** 68 x 49 carrier; provisional CM0, SEC1210, concealed microSD, power.
- **E** Four L92 cells; four pairs of Keystone contacts.
- **F** MJF PA12 rear shell; slide/latch door; 4 x M2 screws.

SERVICE ORDER **Rear shell -> cells -> carrier -> front**

Section views and thickness stack

SECTION A-A / UPPER DISPLAY-CARD-CARRIER STACK

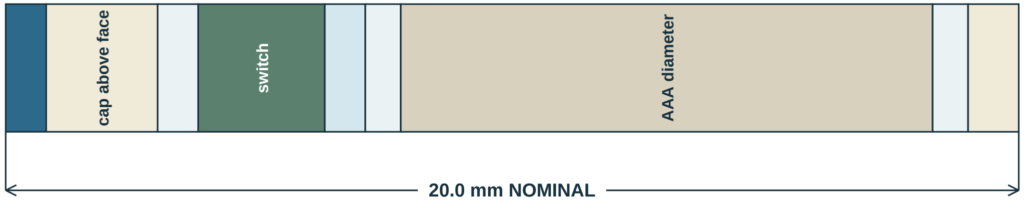


Nominal modeled stack sums to 20.0 mm. Builder must verify actual display header removal, connector solder fillets and shell ribs from supplier STEP release.

1 Z-STACK WARNING

18 mm is not a responsible quotation target with catalog AAA cells, 2.5 mm tactile switches and serviceable shells. The 20 mm envelope has essentially no reserve at the keypad-battery overlap. If measured parts exceed the stack, increase rear bulge locally to 20.8 mm; do not reduce cell compression or switch travel.

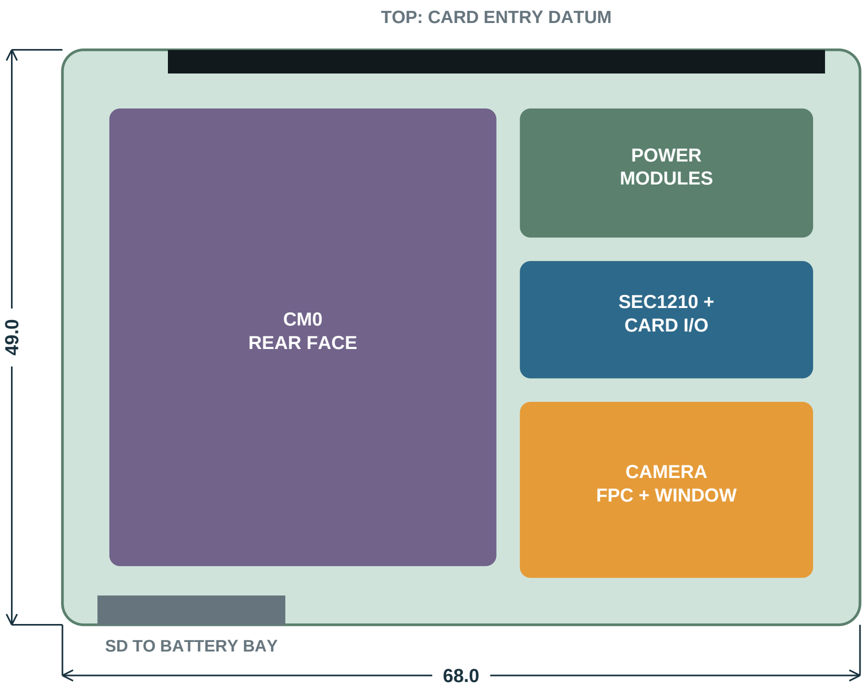
SECTION B-B / LOWER KEYPAD-BATTERY STACK



2 CRITICAL DIMENSIONS

Faceplate flatness 0.20 mm max. Display-lens gap 0.20-0.40 mm. Battery contact compression 0.5-0.8 mm. Camera foam compression 10-20%. Keycap-to-face radial gap 0.20-0.30 mm per side. No component may touch a battery wrapper.

MAIN CARRIER / 68 x 49 x 1.0 mm / 4-LAYER ENIG



1 MAIN-BOARD FABRICATION

4-layer FR-4, 1.0 mm, ENIG, 1 oz copper. L2 solid ground; L3 power/slow signals. CM0 is provisional for P0. Route Camera Module 3 clock plus data lanes 0/1 as controlled 100 ohm differential pairs. Observe the official CM0 land pattern and keep-outs exactly.

2 KEYPAD BOARD

68 x 54 x 0.8 mm, 2-layer ENIG. 20 Alps SKRP tactile switches, 20 neutral-white 0603 LEDs, 19 light wells, MCP23017 I/O expander and a 6-way board cable. Use two switches under the 0 key; parallel them at the matrix inputs so either contact registers zero.

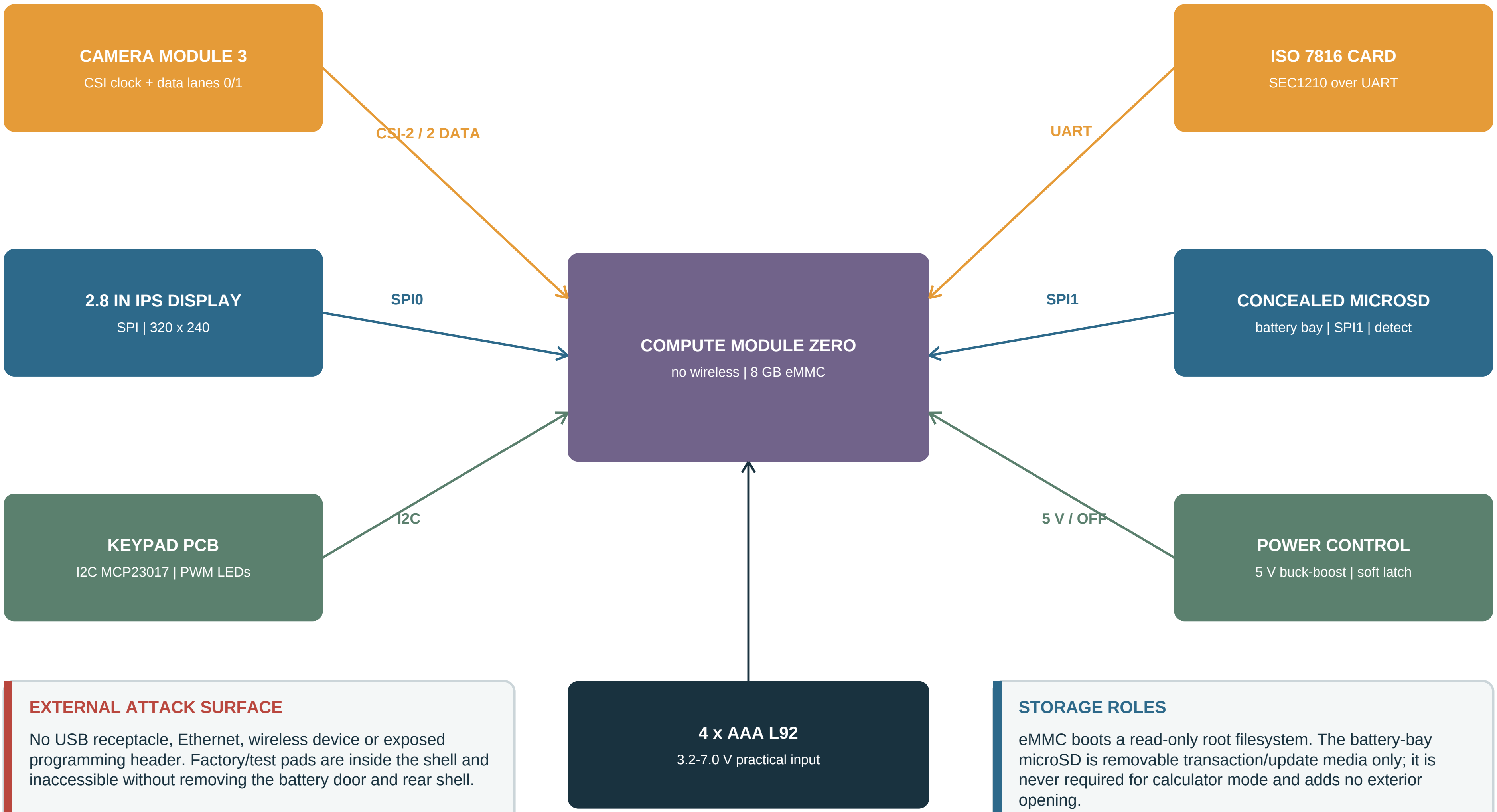
3 DESIGN-FOR-ASSEMBLY

Use 0402 only where the CM0/CSI fan-out requires it; use 0603 elsewhere. All user-edge connectors must have mechanical shell support. No hand wires in the released prototype except replaceable battery-contact leads. Put fiducials and revision marking on both boards.

4 RELEASE DEPENDENCY

ECAD must import supplier STEP models for display, camera, smartcard and power modules. A 1:1 paper/laser-cut fit check precedes PCB order.

Electrical block diagram



EXTERNAL ATTACK SURFACE

No USB receptacle, Ethernet, wireless device or exposed programming header. Factory/test pads are inside the shell and inaccessible without removing the battery door and rear shell.

STORAGE ROLES

eMMC boots a read-only root filesystem. The battery-bay microSD is removable transaction/update media only; it is never required for calculator mode and adds no exterior opening.

Proposed interface and GPIO allocation

This is a routing proposal, not a CM0 pin-numbering authority. ECAD and firmware engineers must verify every alternate function, voltage domain, boot strap and device-tree overlay against the purchased CM0 datasheet before schematic release.

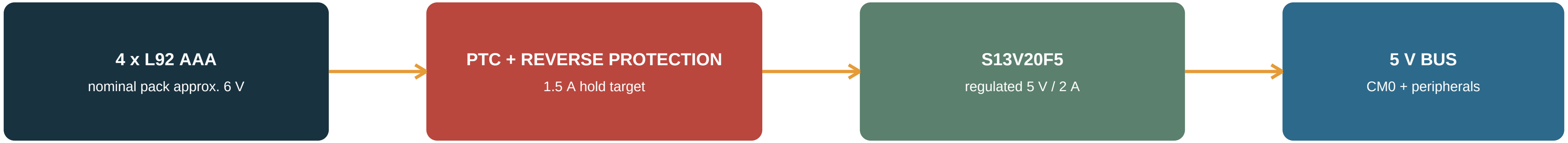
Function	Host interface / proposed GPIO	Peripheral	Electrical notes
Display pixel data	SPI0: GPIO11 SCLK, GPIO10 MOSI, GPIO9 MISO, GPIO8 CE0	ST7789 on Waveshare board	3.3 V logic. Remove Pico headers. MISO retained only if board requires reads.
Display control	GPIO24 D/C; GPIO25 reset; GPIO12 PWM	TFT reset and backlight	Use 1 kohm series on control lines; MOSFET if module backlight load exceeds GPIO.
Key matrix	I2C: GPIO2 SDA, GPIO3 SCL	MCP23017, address 0x20	5 columns + 4 rows; IRQ optional. Use CM0 on-module 1.8 kohm pull-ups; external pull-up footprints DNP.
Key legend LEDs	GPIO13 PWM through MOSFET + card-detect inhibit	20 x white LED + resistors	Hardware-off without card. One wallet-mode brightness zone; target 2-4 mA per LED max.
Smartcard interface	UART0: GPIO14 TX, GPIO15 RX; GPIO27 reset/IRQ	Microchip SEC1210	Follow SEC1210 reference exactly. Card VCC/CLK/RST/I/O handled by the controller.
Transaction microSD	SPI1: GPIO21 SCLK, GPIO20 MOSI, GPIO19 MISO, GPIO16 CS	Hirose DM3AT push-push	Downward entry into battery bay; series damping; card-detect to expander input.
Camera	CM0 4-lane-capable CSI host; route CLK + D0/D1	Camera Module 3 on 15-way FPC	The 15-way camera connector exposes only lanes 0/1. Verify pin mapping against both official schematics.
Power shutdown	GPIO26 active pulse	Pololu Mini Pushbutton Power Switch LV	Firmware asserts OFF after storage sync. Red C/ON switch also connects to power switch ON via diode.
Battery monitor	I2C shared; address 0x48	ADS1015 + switched divider	Divider is enabled only while measuring; calibrate low-battery thresholds on real L92 discharge.
Factory service	Internal pads: 5 V, GND, D+, D-, nRPIBOOT, TX, RX	Pogo fixture only	No external opening. Mark as factory-only. Fit ESD; remove batteries during USB flashing.

FIRMWARE OWNERSHIP

Freeze one signed pin-map CSV, schematic netlist and device-tree overlay together; mismatched revisions are a release blocker.

Power subsystem and shutdown behavior

POLULU MINI PUSHBUTTON SWITCH LV CONTROLS INPUT PATH



CONTROL SEQUENCE

State	User action / condition	Hardware action	Software action
Off	Press red C/ON	Diode-isolated contact pulses ON input	CM0 boots from eMMC; calculator UI appears first.
Running	Short red C	No power state change	Cancel/back or clear according to current UI.
Shutdown request	Hold red C for 2.0 s	Key remains readable by matrix	UI asks confirmation; sync; unmount microSD; stop services.
Orderly off	Firmware reaches safe-off	GPIO26 pulses OFF input	Read-only root remains clean; 5 V bus disconnects.
		rtion	

1 ENERGY EXPECTATION

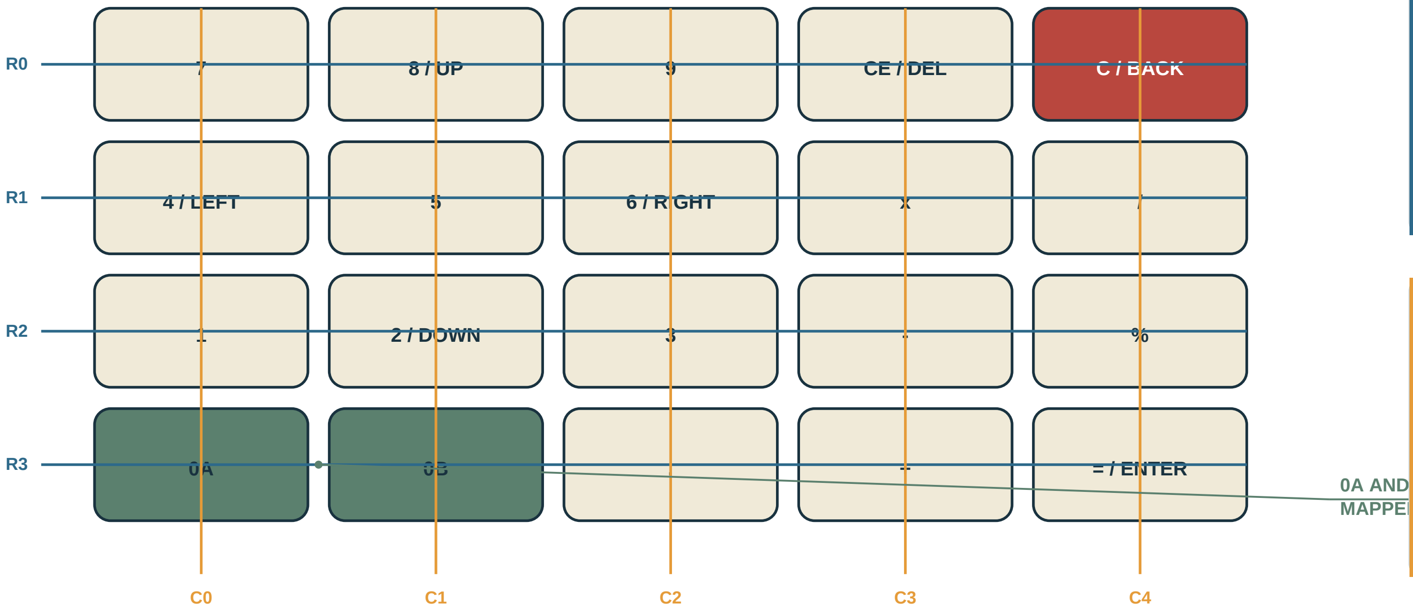
QK2-04 estimates about 7.2 Wh for four L92 cells. With a measured 1.0-1.8 W active load, expect roughly 3-6 h active runtime after converter and cold-temperature losses, plus much longer intermittent calculator use. Treat this as a test target, not a guaranteed specification.

2 PROTECTION AND BULK CAPACITANCE

Fit 470 uF low-ESR polymer plus 22 uF and 0.1 uF at the 5 V rail. Verify startup at depleted cells with display and camera enabled. Qualification includes reverse-cell insertion, one-cell missing, shorted output and abrupt hard-off without heat or wrapper damage.

Display, keypad and backlight implementation

KEY MATRIX / 5 COLUMNS x 4 ROWS



1 PURCHASED INPUT PARTS

Alps Alpine SKRP series tactile switch, nominal 4.2 x 3.2 x 2.5 mm, selected for compactness. Use the exact purchasable force variant confirmed by sample testing. Do not substitute membrane domes for P0; physical confirmation must remain crisp and repeatable.

2 PROTOTYPE KEYCAPS

SLA translucent resin, opaque ivory/red coat, then laser-ablate only secondary wallet legends. Calculator digits and operators remain dark printed marks. A black light-well frame prevents bleed. Wallet legends must be invisible at arm's length while unlit.

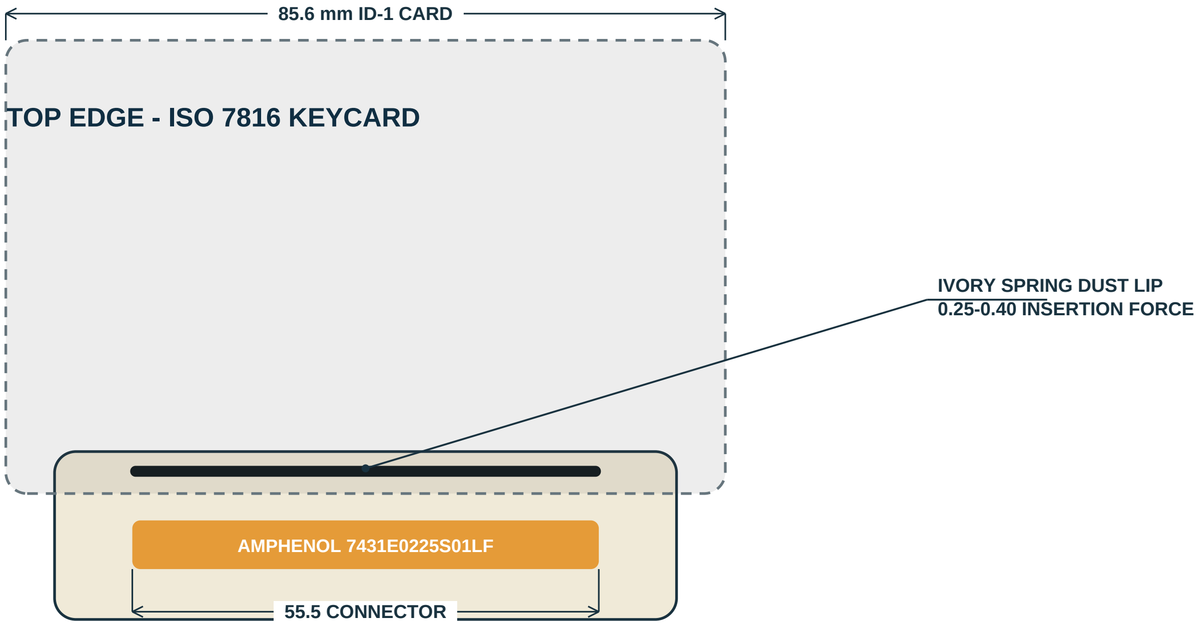
3 DISPLAY MODULE PREPARATION

Use Waveshare Pico-ResTouch-LCD-2.8. Remove the Pico female headers by a rework shop; do not use the resistive touch layer or the module's microSD socket. Connect display only through a low-profile harness to the carrier. Verify module revision before wiring.

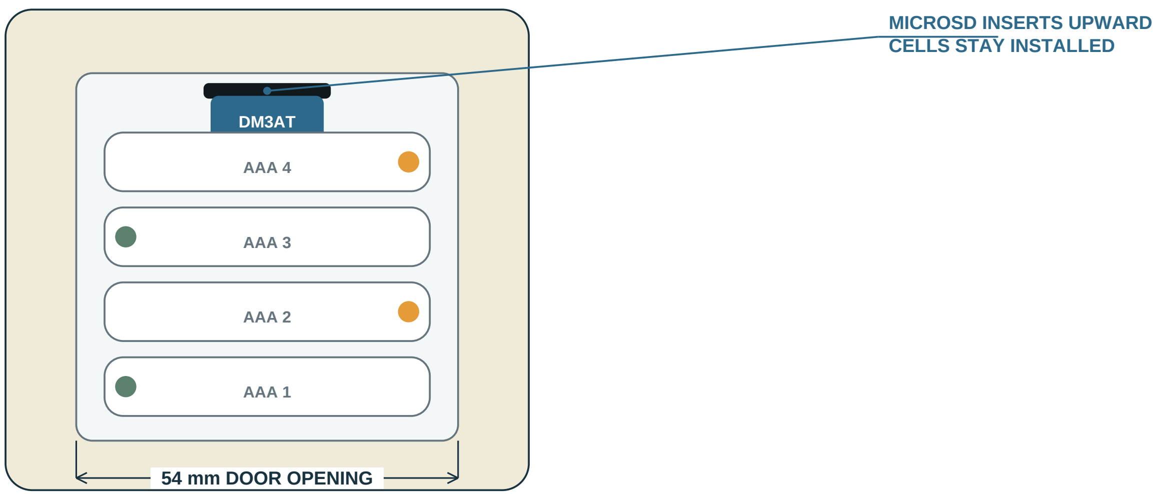
4 OPTICAL AND MODE ACCEPTANCE

With a card present, wallet legends are readable at 30% PWM without glowing key edges. With no card, LEDs remain electrically dark under boot, idle and every keypress, and secondary navigation marks are not identifiable at arm's length. CCT 4000-5000 K.

Smartcard and concealed microSD details



REAR - BATTERY DOOR REMOVED



1 SMARTCARD ASSEMBLY RULE

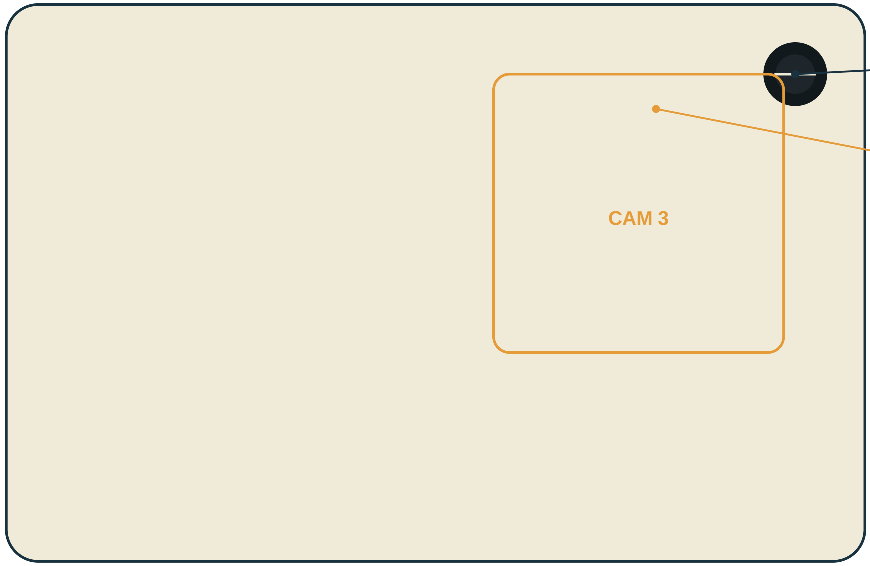
Connector is push-pull, 8 contacts plus card-detect, 55.5 x 40 x 5.8 mm. The shell must support its entrance flange so insertion load does not reach PCB solder joints. Nominal card insertion depth is about 40 mm, leaving about 45.6 mm exposed. Verify contact landing against the actual keycard artwork and chip position.

2 MICROSD ASSEMBLY RULE

Mount Hirose DM3AT-SF-PEJM5 on the carrier at the top of the battery opening, socket mouth facing downward. Keep at least 3.0 mm finger clearance and 1.0 mm shell clearance around an inserted card. Card detect and ESD protection remain required. The battery door, not the cells, conceals the opening.

3 STEALTH / SERVICE RESULT

No sixth exterior opening. The user can exchange transaction/update media after opening one familiar service door without disturbing cell order or exposing the blue faceplate.



5.5 mm FALSE-SCREW RECESS
INTERCHANGEABLE 3.4 / 4.2 / 5

CATALOG AR WINDOW, 5-6 mm
0.4-0.7 mm THICK; BLACK FOA

1 MECHANICAL OPTICAL RULE

Use standard-FoV Camera Module 3. Lens center tolerance is +/-0.25 mm. The recessed anti-reflective window and its foam gasket are part of the frozen optical stack; the bore is never left open. Interior surfaces are matte black. Verify the actual catalog window lot before Gate A.

RECOVERY-DOCUMENT FOOTER TARGET

```
cv0qpzry9x8gf2tvdw0s3jn54khce6mua7lqpzry9x8gf2tv

dw0s3jn54khce6mua7lqpzry9x8gf2tvdw0s3jn54khce6mu

a7lqpzry9x8gf2tvdw0s3jn54khce6mua7lqpzry9x8gf2
```

FORMAT DIAGRAM ONLY - USE FROZEN VECTOR TEST ARTEFACT

2 TOKEN STRUCTURE

Exactly 142 Bech32 characters: cv0 sentinel + 133 data + 6-character BIP-173 checksum, rendered as an approximately 48-character x 3-line monospace block. RS(83,49) recovery is external to the printed token.

3 O01 - GATE A CAPTURE MATRIX

Use the production token artwork across frozen printer, paper, scale, distance, angle, illumination, defocus, motion and degradation points. For every character report correctly read, erasure-marked or silently misread. Also record RS errors, erasures and final token recovery.

4 O02 - APERTURE / WINDOW FREEZE

Derive the confidence floor and top-two probability margin from the predeclared dataset. Select the smallest aperture-and-window combination with zero observed silent misreads and acceptable erasure/recovery performance. A pass/fail QR decode result is not an acceptable substitute.

Enclosure, materials and drawing tolerances

Part	Prototype process / material	Nominal	Finish / color	Critical notes
Front shell	MJF PA12, bead blasted	1.2-1.5 mm walls	Ivory, low gloss	Heat-set M2 inserts only in bosses >=5.0 mm OD. Black light-fence insert locates keypad.
Rear shell	MJF PA12, bead blasted	1.2-1.5 mm walls	Ivory, low gloss	Mold/print four AAA channels. Camera aperture boss upper-right. No unsupported wall >35 mm.
Faceplate	Laser-cut 5052-H32 aluminum	0.8 mm	Blue anodize or fine texture powder coat	Flatness <=0.20 mm; deburr all key windows; removable with front carrier.
Display lens	Laser-cut smoke PMMA	0.5 mm	25-35% visible transmission	Black perimeter print/tape; no adhesive in active area.
Keycaps	SLA translucent resin	10.7 square; 2.2 high	Ivory; red C	Opaque coat, laser-ablate legends. 0 key 23.7 x 10.7.
Battery door	MJF PA12	1.0 mm skin + ribs	Ivory	Slide down 2 mm then lift; captive latch; 0.5-0.8 mm cell pad compression.
Fasteners	Catalog stainless Torx	4 x M2 x 6	Black or dark zinc	Two top, two bottom. Upper-right exterior boss is optical faux screw, not a structural screw.
Optical window	Catalog AR-coated glass	5-6 mm dia; 0.4-0.7 mm	Clear, low reflection	Recess behind faux screw. Qualify the purchased lot in O01/O02 before aperture freeze.
Camera seal	Die-cut black PORON/EPDM	0.8 mm	Matte black	Seal around window and camera pocket; 10-20% assembled compression.

1 GENERAL TOLERANCE

Unless noted: linear +/-0.25 mm for machined/laser parts, +/-0.40 mm for MJF, angles +/-1 deg. Apply 0.20-0.30 mm per-side clearance to moving printed parts after a fit coupon.

2 CORNER AND EDGE

External radius 6.5 mm. Internal radii >=1.0 mm. All hand-contact edges R0.4 minimum. Slot edges chamfer 0.25 x 45 deg. No sharp aluminum perimeter exposed.

3 COSMETIC DATUM

Blue face remains visually uninterrupted. The card entrance is in the ivory top edge; microSD is behind the battery door. Reserve a blank compliance area inside the door, but apply no CE, UKCA or FCC mark to P0.

Electrical bill of materials

Ref / qty	Preferred part	Function	Budget EUR	Procurement / substitution rule
U1 / 1	Raspberry Pi Compute Module Zero, CM0000008	No-wireless, 512 MB, 8 GB eMMC	65-85	P0 provisional. Buy/verify exact SKU; measure RAM headroom. No wireless substitute.
DISP1 / 1	Waveshare Pico-ResTouch-LCD-2.8	320 x 240 IPS display	22-30	Prototype-only artifact. Remove Pico headers; do not use touch or onboard SD.
CAM1 / 1	Raspberry Pi Camera Module 3 standard	Autofocus footer-token camera	30-38	Standard FoV; 15-way connector routes clock plus data lanes 0/1.
J1 / 1	Amphenol 7431E0225S01LF	ISO 7816 card connector	6-12	Right-angle, push-pull, card detect. No top-entry replacement.
U2 / 1	Microchip SEC1210-CN-TR	ISO 7816 interface	5-10	Follow Microchip / MIKROE-5492 reference. Verify firmware driver before PCB.
J2 / 1	Hirose DM3AT-SF-PEJM5	Internal microSD push-push	2-5	Battery-bay mouth; ESD and card detect required.
U3 / 1	Pololu S13V20F5 #4085	5 V / 2 A buck-boost	16-22	Solder as module for P0; custom regulator only after measurements.
U4 / 1	Pololu Mini Pushbutton Power Switch LV #2808	Soft power / reverse protection	5-9	Use ON/OFF control pins; preserve firmware orderly-off flow.
U5 / 1	MCP23017	Key matrix I/O	2-4	3.3 V; I2C address 0x20.
U6 / 1	ADS1015	Battery voltage ADC	3-6	Switched divider to reduce off-state drain.
SW1-20 / 20	Alps Alpine SKRP series, force TBD by sample	Tactile keys	8-16	4.2 x 3.2 x 2.5 mm envelope; buy two force variants for test.
LED1-20 / 20	0603 neutral white, 4000-5000 K	Legend light	2-5	Bin same reel. Set resistor after optical coupon.
BAT / 4	Energizer L92 AAA lithium	Removable power	10-16	Binding chemistry for P0 qualification.
CONTACT / 8	Keystone AAA spring/button contacts	Battery interconnect	4-8	Choose paired catalog parts after shell coupon.
WIN1 / 1	Catalog AR-coated round optical window	Sealed camera cover	5-15	5-6 mm dia, 0.4-0.7 mm. Freeze supplier lot with O01/O02.
MISC / lot	ESD, PTC, MOSFET, caps, FPCs, headers, passives	Protection/interconnect	25-45	0603 preferred; 470 uF polymer at 5 V bus.

PARTS SUBTOTAL **approximately EUR 210-330, excluding PCBs, enclosure, assembly, spares and tax**

Fabrication budget and procurement package

Lot	P0 quantity / process	Budget EUR	Builder quotation basis
Main carrier PCBs	5 bare, 4-layer, 1.0 mm ENIG; 2 assembled	70-140	Controlled impedance on CSI; CM0 and fine-pitch assembly priced separately if required.
Keypad PCBs	5 bare, 2-layer, 0.8 mm ENIG; 2 assembled	35-80	20 switches and LEDs per board; include electrical test.
MJF shell set	2 front, 2 rear, 2 doors, light fences	80-180	PA12, bead blast; one raw fit set before dyed/painted cosmetic set.
Faceplate + lenses	2 laser-cut aluminum; 2 smoke PMMA	40-100	Blue finish after fit confirmation. Deburr and protect cosmetic face.
SLA keycaps	2 sets plus force/legend coupons	35-90	Translucent resin, paint and laser ablation. Include double-width 0.
Electrical parts	Two device sets + selected spares	330-520	Includes CM0 market pricing, window, connectors, passives and four L92 cells per build.
Assembly fixtures	Pogo programming jig; card/cell fit gauges	50-150	Reusable. No external service connector added to product.
Expected P0 material/fab	Two units, no engineering labor	640-1,260	Wide range reflects one-off European service pricing and component availability.
Turn-key prototype service	CAD, ECAD, firmware bring-up, assembly, test	2,500-6,000	Indicative only. Quote to milestones below, not one undifferentiated fixed-price build.

REQUIRED SUPPLIER DATA PACK

- Official dimension drawing and STEP model for every purchased module; record manufacturer revision and supplier URL.
- One approved AVL line per critical part: compute, display, camera, optical window, smartcard connector, microSD and power modules.
- No substitution of display board, CM0 SKU, camera FoV, cell chemistry or card connector without written deviation approval.
- Buy one spare display, camera and CM0 for a two-unit run; connector and switch attrition minimum 10 percent.

Prototype assembly sequence

01
FIT COUPONS
Print battery channel, key well, card entrance, battery-bay microSD and sealed-window coupons. Confirm cell compression, cap gap, connector support and service access before full shells.

02
MODULE BRING-UP
On the purchased CM0, prove eMMC boot, SPI display, Camera 3 footer-token frames, SEC1210 with real card, internal SPI1 microSD and soft-power off. Run P04 worst-case multi-frame RAM test before ECAD.

03
PCB RELEASE
Freeze schematic, pin-map CSV, device tree and supplier revisions together. Run DRC, impedance review, land-pattern review and 1:1 print fit.

04
PCB ASSEMBLY
Assemble one carrier and one keypad first. AOI and continuity check before fitting CM0/power modules. Program eMMC through internal pogo fixture with cells removed.

05
FRONT SUBASSEMBLY
Fit faceplate and display lens; install light fence, keys and keypad PCB. Verify every cap returns freely at four corner presses and no LED bleed is visible.

06
UPPER ELECTRONICS
Install display, card connector/carrier, Camera 3, qualified AR window and foam seals. Verify card detect, window compression and optical center before joining shells.

07
BATTERY SYSTEM
Install contacts and insulation barriers. Polarity-mark inside door. Test each channel for spring force and no metal burrs; then fit four fresh L92 cells.

08
FINAL CLOSE
Route FPCs without creases; confirm no battery contact with solder. Torque four M2 screws to 0.12-0.16 N m. Apply internal serialized prototype label.

09
ACCEPTANCE
Run the matrices on pages 19-20, document deviations, photograph interior and archive as-built BOM, firmware hash and PCB/enclosure revision.

STOP-WORK RULE

Any pinched cell, hot part, card-contact mismatch or uncommanded shutdown stops assembly.

Mechanical, power and usability acceptance

ID	Test	Method / condition	Pass criterion	Gate
M01	Envelope and mass	Caliper all axes; scale with 4 x L92 and no card	112.0 x 74.0 x <=20.0 mm within stated tolerances; 165-195 g target	Build
M02	Key freedom	Press center and four corners, 100 cycles each	No bind; all presses register once; 0 works on either underlying switch	Build
M03	Card insertion	20 insert/remove cycles with production-format keycard	Chip contacts align; detect reliable; 45.6 +/-2 mm card remains exposed	Build
M03a	Dust-shutter life	500 full card insertion/removal cycles; inspect every 100	Shutter closes flush, no binding/debris; card detect and contacts remain reliable	Build
M04	microSD service	20 exchanges through open battery door, cells installed	No cell removal or tool required; no ejection into bay; card detect reliable	Build
P01	Power startup	Fresh and depleted-cell simulator; camera/display on	No reset loop; 5 V within module limits; no part >60 C in 25 C ambient	Build
P02	Orderly shutdown	50 software offs and 10 recovery hard-offs	No eMMC corruption; microSD cleanly unmounted on every orderly off	Build
P03	Battery faults	Reverse one cell, omit one cell, output short through fixture	No smoke, wrapper heat or permanent unsafe failure; recovery after correct cells	Build
U01	Display completeness	Longest supported address and review transcript	No address truncation; full value and destination review accessible before confirm	Gate A
U02	Legend lighting	No-card boot/idle/keypress; card-present 30% PWM	No-card LEDs electrically dark; wallet marks invisible at arm's length; no edge glow or green tint when lit	Gate A
U03	Dice digit entry	Enter 99 sequential digits at normal and fast cadence	No missed or doubled keystroke; each digit receives immediate visual confirmation	Gate A
R01	Calculator decoy	Boot without keycard; user test	Normal calculator is first and complete; wallet surface not advertised	Gate A

RELEASE CONDITION **P0 may be demonstrated with test keys only after all Build rows pass. Real funds remain prohibited until independent Gates A-C are signed.**

Optical, memory and security acceptance

ID	Test	Method / condition	Pass criterion	Gate
O01	Footer-token OCR	Frozen 142-character token across complete print, capture and degradation matrix	Per-character correct / erasure / silent-misread report; confidence floor and top-two margin frozen; zero observed silent misreads in declared set	Gate A
O02	Aperture + window	Compare 3.4, 4.2 and 5.0 mm inserts with production optical window	Smallest stack satisfying O01 with acceptable erasure and RS-recovery performance selected	Gate A
P04	CM0 RAM headroom	Worst-case multi-frame capture, OCR confidence classification and RS recovery on actual CM0000008	Peak RSS, MemAvailable, CMA/buffer use and frame drops recorded; no OOM, swap or dropped required frame; release threshold signed before ECAD	Release
S01	No radios / host ports	Visual BOM audit, OS enumeration, RF sanity scan	No radio silicon, USB receptacle or externally accessible programming pads	Gate C
S02	Secret-at-rest posture	Inspect eMMC and microSD after powered workflows	No terminal wallet secret persisted; logs and swap policy match QK2-04	Gate C
S03	Power-loss recovery	Cut pack during boot, card read, footer capture and update	Returns to calculator or controlled recovery; no unsafe partial-update state	Gate C
C01	Window contamination	Dust exposure, drawer aging simulation and cleanability check	No ingress to lens pocket; user cleaning reaches only exterior window; O01 still passes	Build

1 O01 REPORTING RULE

A silently wrong character spends two RS parity symbols; an erasure spends one. Therefore confidence thresholds optimize for zero observed silent misreads, not maximum raw character acceptance. Archive the full confusion table, not only final-token pass/fail.

2 P04 RELEASE RULE

P04 is measured during stage 02 bring-up and blocks PCB release. The resulting controlled value must include the exact frame count, resolution, pixel format, OCR build, kernel/CMA settings and minimum remaining memory. Write a one-page successor migration note in parallel.

Open items, risks and builder deliverables

BLOCKERS BEFORE PCB / SHELL RELEASE

R1 — **CM0 PROCUREMENT / MEMORY**

CM0000008 is the required P0 SKU, but procurement and 512 MB RAM remain risks. Buy samples, complete P04 and issue a one-page successor migration note before ECAD.

R2 — **SEC1210 DRIVER**

Prove actual keycard ATR, APDU exchange and voltage class using the Microchip/MIKROE reference circuit on bench.

R3 — **20 MM LOWER STACK**

Measure caps, switch, keypad PCB, insulation, real L92 diameter and door ribs. Allow local 20.8 mm rear bulge only by deviation.

R4 — **P0-ONLY DISPLAY MODULE**

Purchase sample and capture as-built geometry. Touch layer, unused SD socket and manual header rework are accepted only for P0 and shall not migrate forward.

R5 — **SEALED CAMERA STACK**

Freeze aperture, AR-window lot and gasket only after the footer-token O01/O02 matrix; faux-screw aesthetics do not outrank silent-misread control.

R6 — **QK2 DOCUMENT CONTROL**

QK2-05 or signed deviations must adopt the compact 112 x 74 x 20 envelope, 320 x 240 display, top card entrance and card-present lighting policy.

THIRD-PARTY BUILDER SHALL RETURN

- Native mechanical CAD plus STEP and dimensioned PDF drawings for all enclosure parts and keycaps.
- Schematic PDF, source ECAD, pin-map CSV, Gerber/ODB++, drill, centroid, assembly drawings and manufacturing BOM with MPNs.
- Pogo-fixture drawing, programming procedure, firmware image/hash, device-tree source and reproducible build note.
- As-built inspection report: dimensions, mass, current, temperature, optical result, key test and all substitutions/deviations.
- Interior/exterior photographs and serialized configuration record for each prototype unit.
- CM0 procurement evidence, P04 memory report, one-page successor migration note and review-reconciliation record.

CONFIGURATION RULE

A hardware change is incomplete until schematic, PCB, enclosure, firmware, test plan and QK2 deviation log all point to the same revision.

Reference parts and source links

Canonical basis	QuietKey QK2-04 System & Device Blueprint, user-supplied PDF, effective 15 Aug 2026.
Historical aesthetic	https://americanhistory.si.edu/collections/object/nmah_334508
Raspberry Pi Compute Module Zero	https://www.raspberrypi.com/products/compute-module-zero/
Official CM0 datasheet RP-009251	https://pip.raspberrypi.com/documents/RP-009251-DS-compute-module-zero-datasheet.pdf
Raspberry Pi LPDDR2 supply statement, Apr 2026	https://www.raspberrypi.com/news/a-new-3gb-raspberry-pi-4-for-83-75-and-more-memory-driven-price-increases/
CM0000008 distributor price / stock snapshot	https://www.lcsc.com/product-detail/C52108793.html
Raspberry Pi Camera Module 3	https://www.raspberrypi.com/products/camera-module-3/
Camera Module 3 reference schematic RP-008969	https://pip-assets.raspberrypi.com/categories/1207-design-files/documents/RP-008969-SD-3-Raspberry%20Pi%20Camera%20Module%203%20Reference%20Schematic.pdf
Waveshare Pico-ResTouch-LCD-2.8	https://www.waveshare.com/pico-restouch-lcd-2.8.htm
Amphenol smartcard connector 7431E0225S01LF	https://www.amphenol-cs.com/product/7431e0225s01lf.html
Microchip SEC1210	https://www.microchip.com/en-us/product/sec1210
Microchip / MIKROE Smart Card 2 Click reference	https://www.microchip.com/en-us/development-tool/MIKROE-5492
Hirose DM3AT microSD family	https://www.digikey.com/en/product-highlight/h/hirose/dm3at-series
Pololu 5 V buck-boost S13V20F5 #4085	https://www.pololu.com/product/4085
Pololu Mini Pushbutton Power Switch LV #2808	https://www.pololu.com/product/2808/specs
Alps Alpine SKRP tactile switch family	https://tech.alpsalpine.com/e/products/detail/SKRPASE010/
Keystone AAA contacts	https://www.keyelco.com/category.cfm/AAA-Cell-Contacts/Thru-Hole-Mount-AAA-Battery-Contacts/id/1216
EU CE marking guidance	https://europa.eu/youreurope/business/product-requirements/labels-markings/ce-marking/indexamp_en.htm

SOURCE CONTROL

URLs are selection references, not frozen specifications. The builder must download and archive the exact datasheet/drawing revision for each purchased lot, then resolve any conflict in favor of the manufacturer drawing and record the deviation.